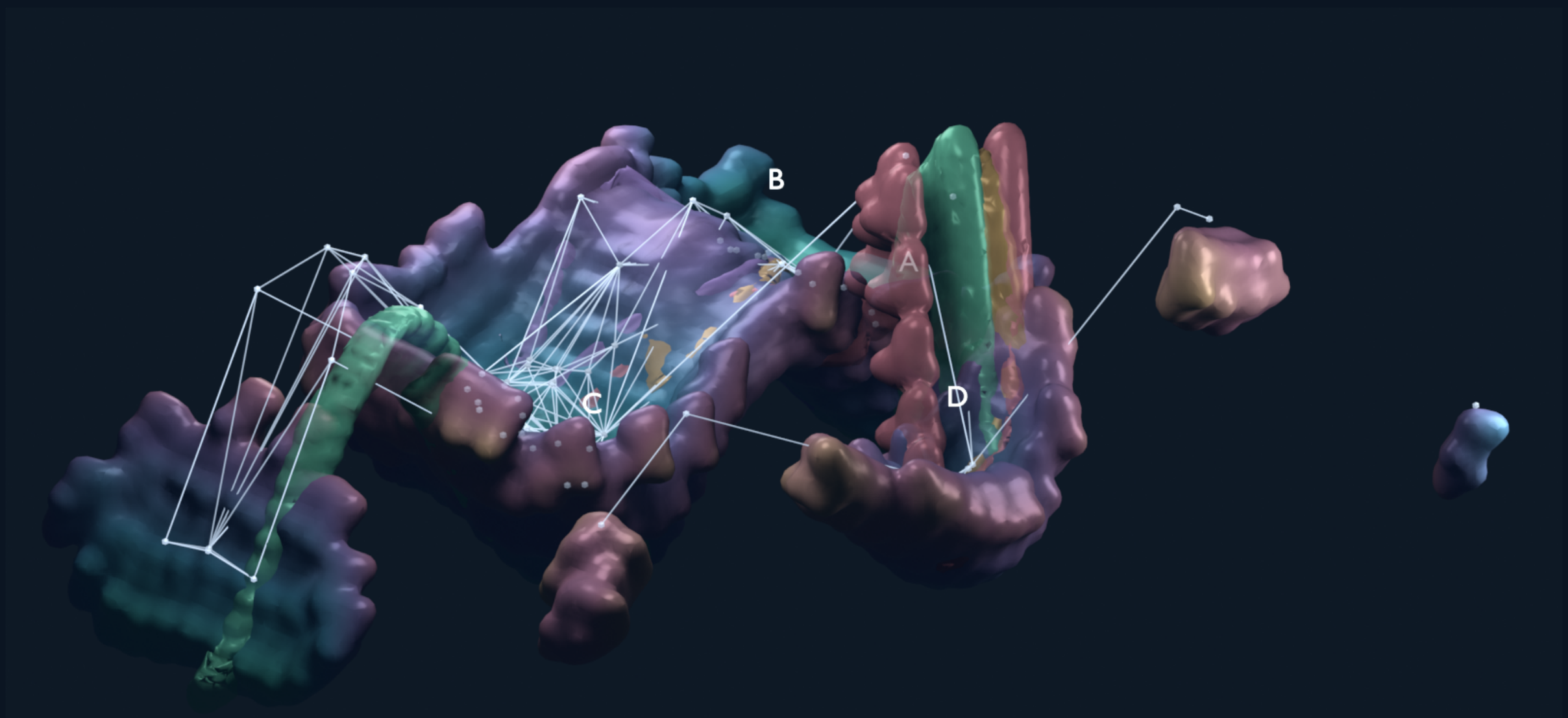
What if a landscape were made of molecules?

And the composition of transitions were something we could see?



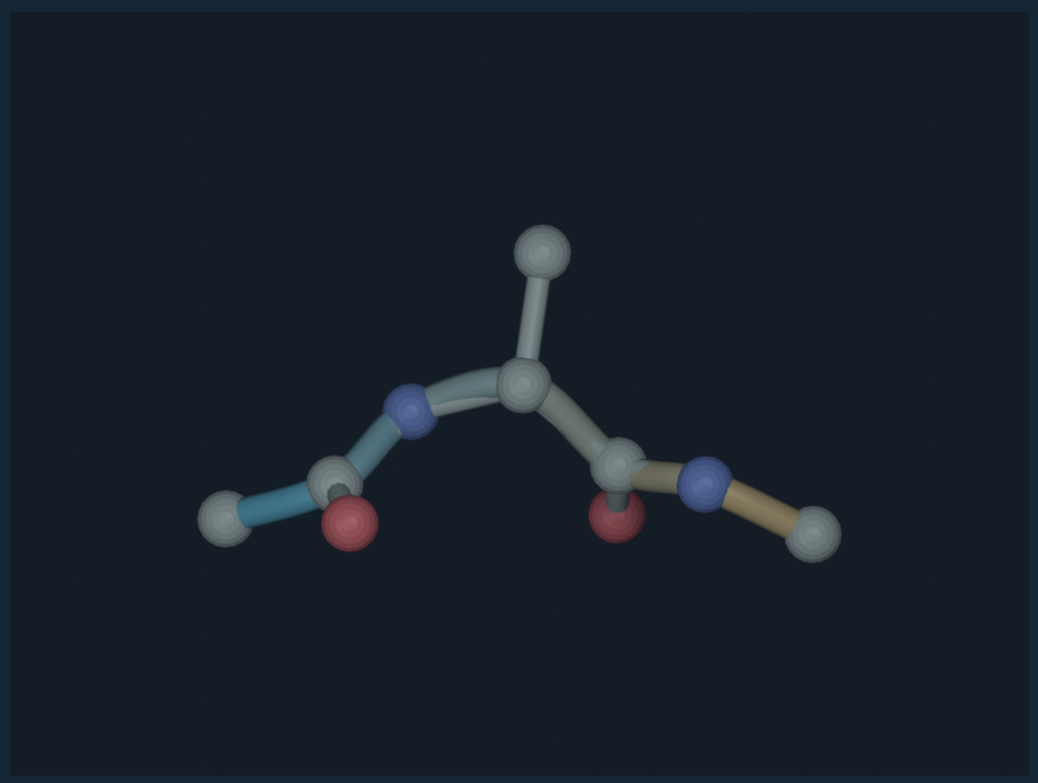
ALANINE DIPEPTIDE + CHIGNOLIN CLN025

Recorded MD geometry. Deforming skins. Simplicial nerves.

Joseph Campagna, PSM

MEET THE MOLECULAR EXAMPLES

Two systems. Different complexity.



ALANINE DIPEPTIDE

A compact torsional benchmark.

Backbone trace with heavy-atom context.





CHIGNOLIN CLN025 / 10 RESIDUES

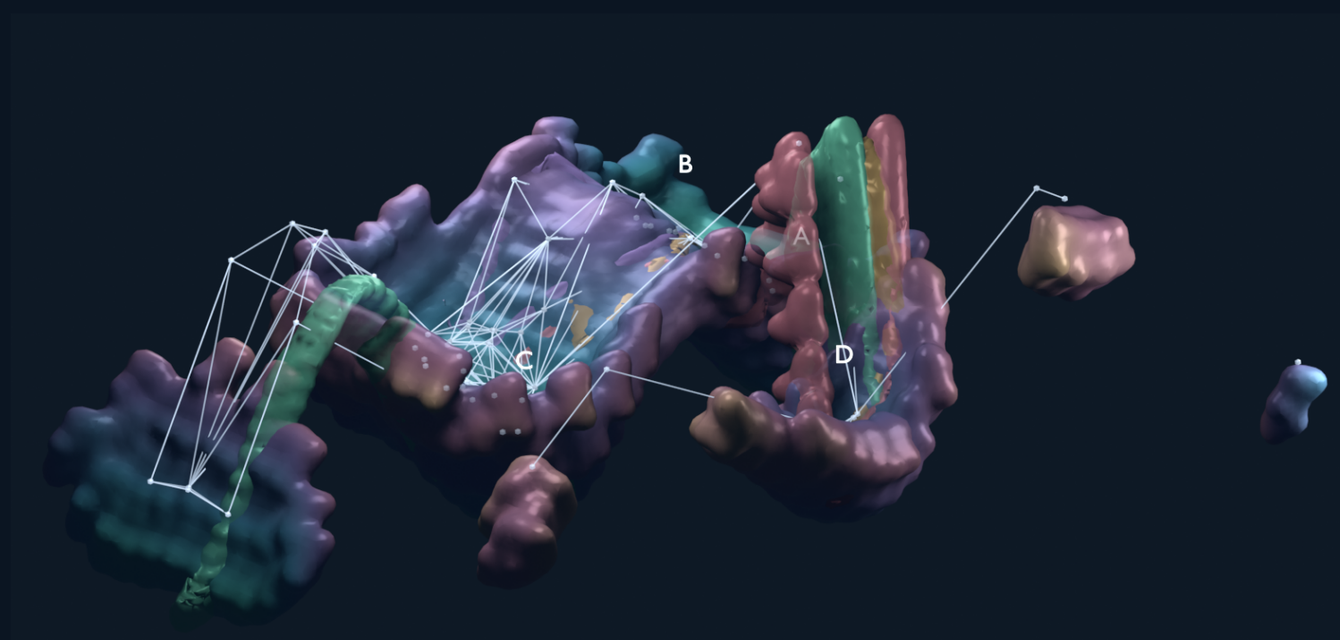
Compact and extended recorded Cα conformations

Cartoons show backbone geometry, not assigned helices or sheets.

FROM RECORDED CONFORMATIONS TO A MOLECULAR ATLAS

Geometry becomes the landscape.

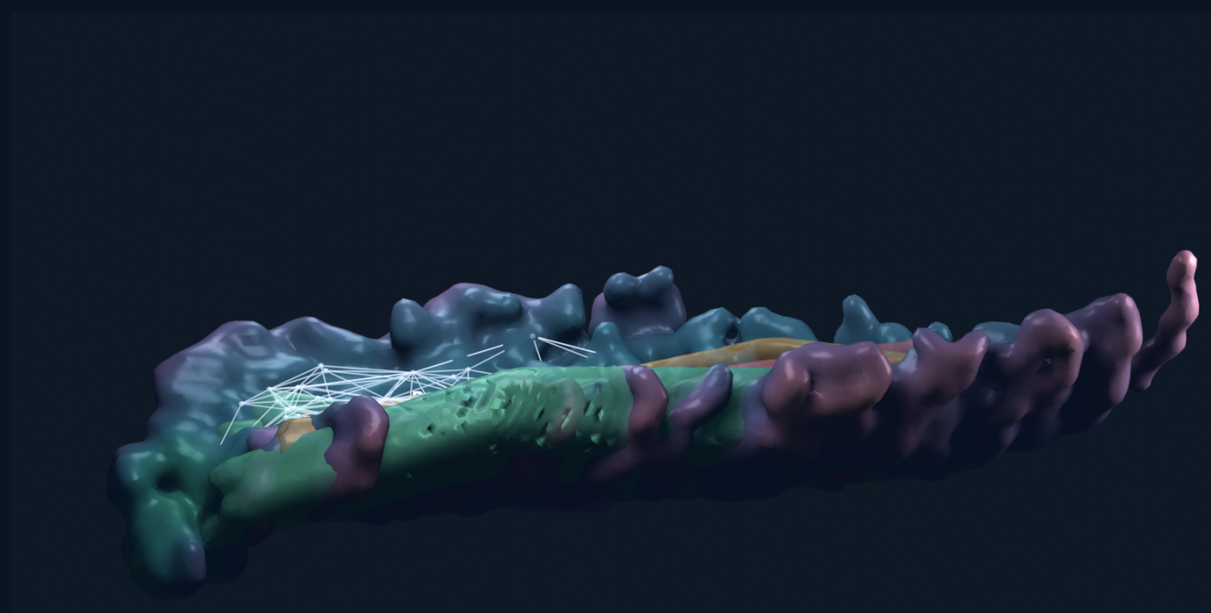
ALANINE / 750,000 FRAMES



Backbone angles

φ and ψ define the chart.

CLN025 / 63,999 FRAMES



Molecular shape

Cα radius of gyration and end-to-end distance.

CLN025 color/height encodes pooled occupancy, not equilibrium free energy.

The geometry moves. Not just the highlights.

Three visual phases, with camera and lighting held fixed.

ALANINE / A → C



PHASE 0 | PHASE ⅓ | PHASE ⅔

CLN025 / A → C

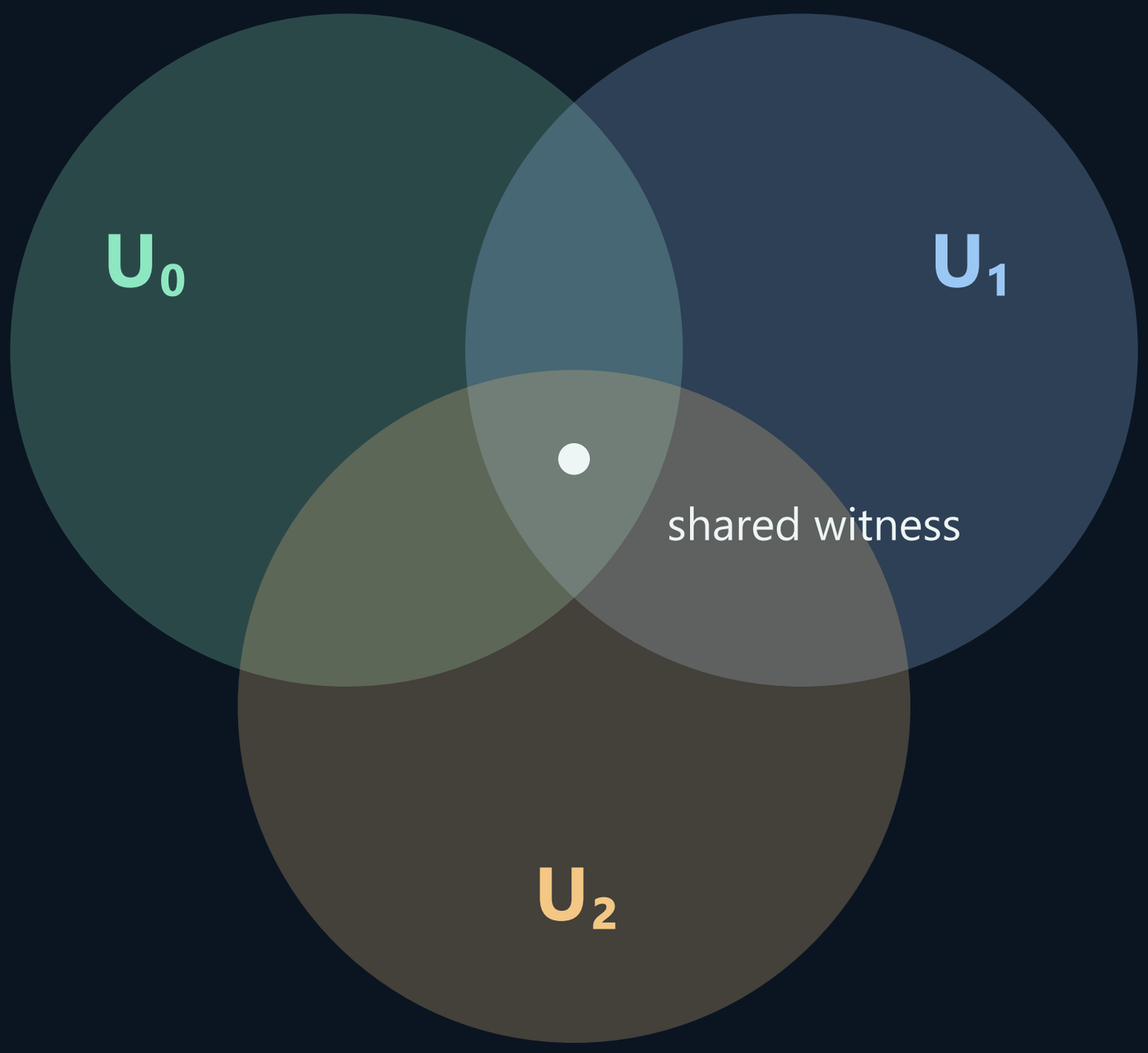


PHASE 0 | PHASE ⅓ | PHASE ⅔

Mesh vertices follow an MD-attached deformation field. These are visual phases, not three instantaneous MD structures.

Which neighborhoods overlap?

A sampled conformation can witness membership in several neighborhoods.



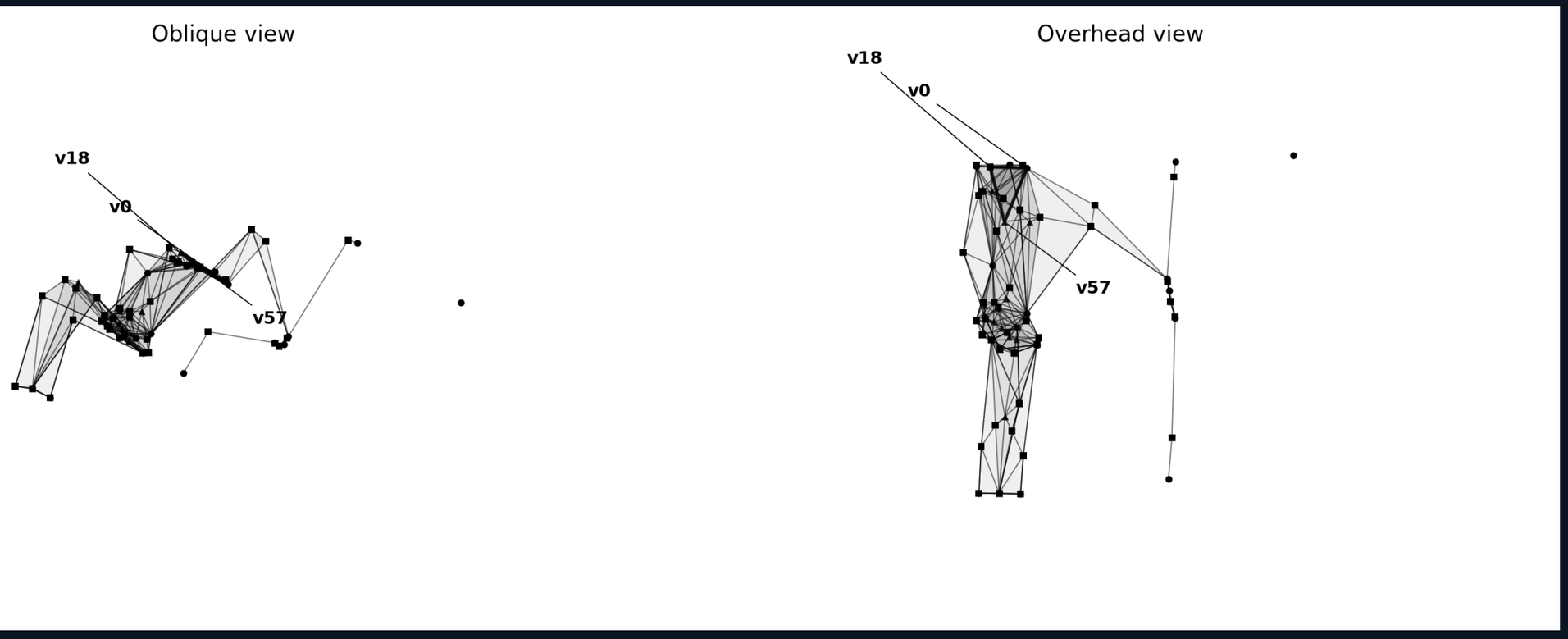
{0} → {0, 1} → {0, 1, 2}

An inclusion chain is a 2-simplex in the face-poset nerve.

Schematic cover. No configuration-space homotopy recovery is claimed.

Alanine dipeptide
The shape of overlap.

91 vertices · 282 edges · 204 faces



LEGEND

Vertex markers: circle = one region; square = pair overlap; triangle = triple overlap.

Edges = index-set inclusions. Shaded faces = composable inclusion chains.

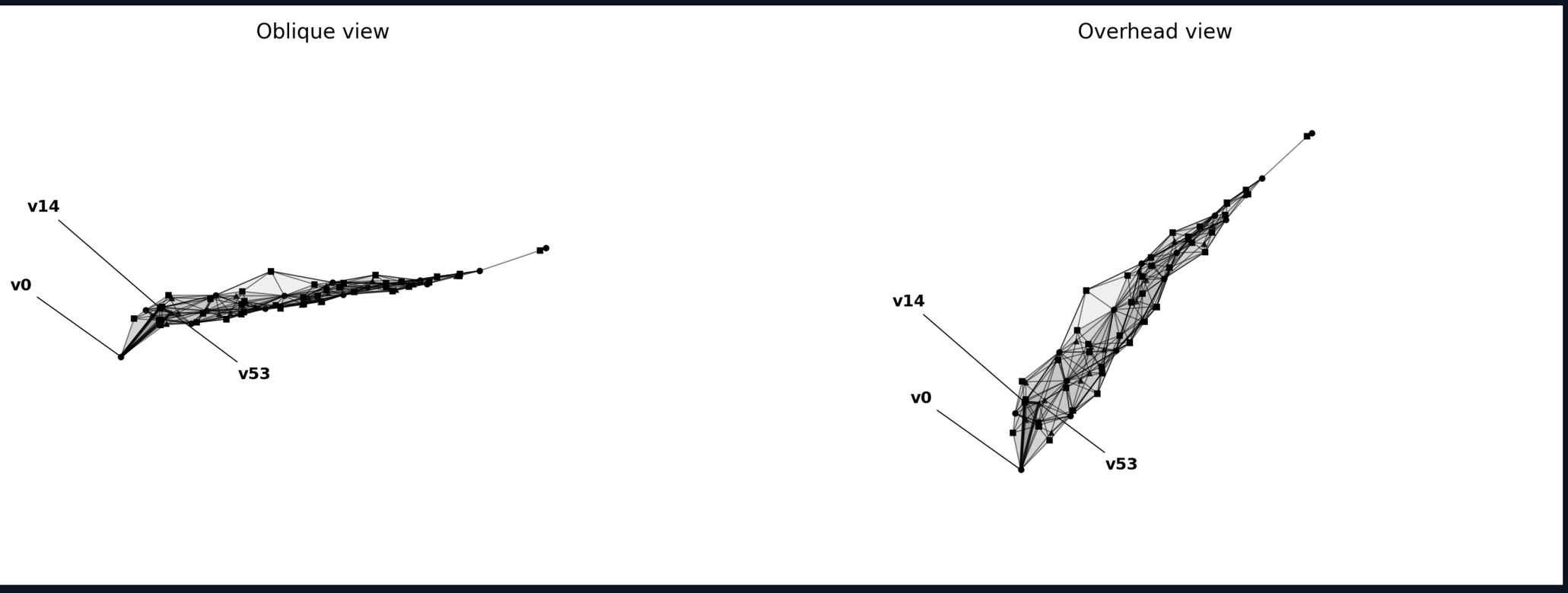
v0 = {0} → v18 = {0,8} → v57 = {0,8,9}

The same 2-skeleton in oblique and overhead views. Bold edges mark the labeled simplex.

Chart: backbone angles φ, ψ. Height: scaled free energy.

Chignolin CLN025
The shape of overlap.

97 vertices · 342 edges · 264 faces



LEGEND

Vertex markers: circle = one region; square = pair overlap; triangle = triple overlap.

Edges = index-set inclusions. Shaded faces = composable inclusion chains.

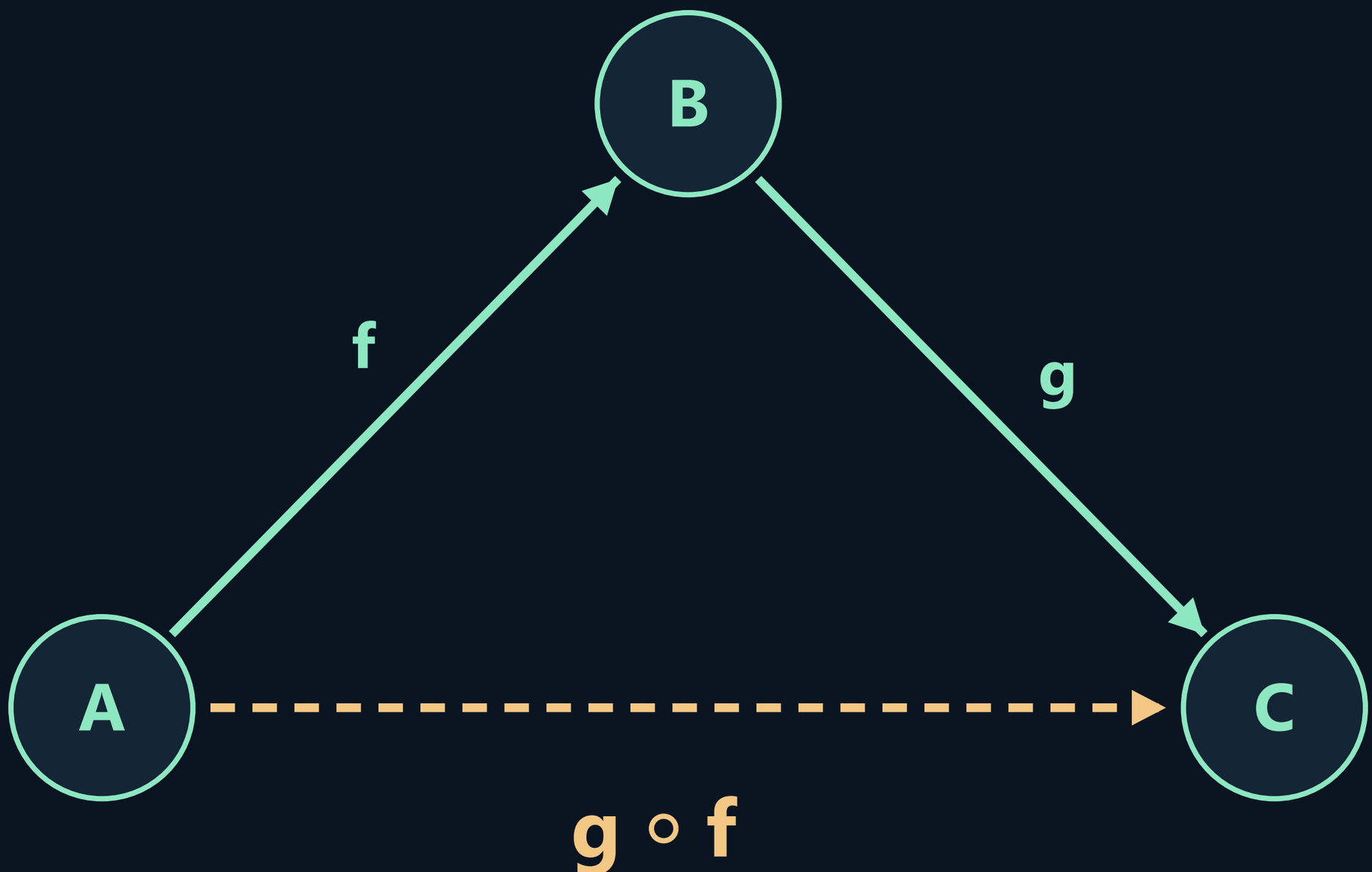
v0 = {0} → v14 = {0,5} → v53 = {0,5,7}

The same 2-skeleton in oblique and overhead views. Bold edges mark the labeled simplex.

Height: pooled occupancy, not equilibrium free energy.

A path can compose.
A shortcut is different.

Observed route types generate a category of paths.



A → B → C need not be a directly observed A → C transition.

Matching endpoints permits formal composition, not a spliced MD trajectory.

A visual experiment.
A precise distinction.

IMPLEMENTED

MD-derived molecular geometry, deforming surfaces, and inspectable overlap/path nerves.

NOT YET VALIDATED

Equilibrium free energies, folding kinetics, or composition of complete dynamical systems.

LIMITATIONS ARE EXPLICIT

Adaptive sampling, projection loss, interpolation artifacts, and conflicting CLN025 timing metadata.

The paper documents the methods, obstacles and future directions.

AN INVITATION TO DISCUSS

How should molecular dynamical models compose?

What should the interfaces be?

Contacts? Torsions? Boundary conditions?

What evidence would show that composition preserves meaningful behavior?

EXPLORE THE VIEWER

joecampdb.github.io/pronerv_MTNerv/

CODE + REPRODUCTION

github.com/joecampdb/pronerv_MTNerv

ILLUSTRATED PAPER

joecampdb.github.io/pronerv_MTNerv/paper.pdf

DATA: MDshare · Pérez Culubret & De Fabritiis (CC BY 4.0)

Sources and scientific limitations are linked in the paper.